

Course Syllabus: Full Stack Web and Generative AI Development (Summer 2026)

Instructor: Phaham | Institution: PEP Academy | Department: Computer Science

Course Overview: 6-week intensive course on modern backend engineering and Generative AI applications.

Week 1: Node.js Core, event loop, libuv, native fs and http modules, asynchronous programming with async/await.

Week 2: Express.js Architecture, routing and middleware, route parameters vs query parameters, and custom logger middleware.

Week 3: Database Design with MongoDB, Mongoose ORM, CRUD operations, aggregation pipelines, JWT auth, and RBAC.

Week 4: Generative AI, LLMs, Google Gemini text-embedding-004, cosine similarity, document chunking, and in-memory vector search.

Week 5: AI Security and Prompt Injection Defense Mechanisms: Layer 1 Query Sanitization, Layer 2 Context Isolation, Layer 3 Output Filtering.

Week 6: Capstone Project Deployment, microservices vs monolithic architecture, environment config, and evaluation metrics.